

CAPM-based portfolio optimisation

Rolling estimation, constrained allocation and risk evaluation

A walk-forward comparison of CAPM-implied and historical-mean portfolio allocations, with trading constraints, transaction costs and explicit analysis of downside risk.

Methods	Python; rolling CAPM; Ledoit–Wolf covariance; constrained optimisation
Data	US sector ETFs and SPY; January 2007–December 2024
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1 Executive summary

This project compares constrained mean–variance portfolios across eleven US sector ETFs. The central design choice replaces historical mean–return estimates with rolling CAPM betas and a fixed 5.5% annual equity risk premium. Ledoit–Wolf covariance shrinkage, monthly rebalancing, position limits and transaction costs complete the allocation framework.

The base run produces 178 monthly return observations from March 2010 to December 2024. CAPM-MV has an annualised return-to-volatility ratio of 0.739, compared with 0.819 for historical-mean Naive MV and 0.881 for SPY.

The comparison reveals a trade-off. CAPM-MV has a less severe maximum drawdown than Naive MV (−25.7% versus −30.7%), but higher historical one-month 95% Expected Shortfall (9.99% versus 9.18%). It leads Naive MV on the reported ratio in 2020–2024 and trails it in the earlier sub-periods.

A 24-month lookback gives a ratio of 0.897 in the historical sensitivity analysis. Varying the lookback and risk-aversion parameter shows how estimation choices affect the balance between return, drawdown and monthly tail risk.

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2 Data and scope

2.1 Objective and benchmarks

Mean–variance optimisation can be sensitive to estimated expected returns [1, 8]. This project examines a CAPM-based return structure alongside historical sample means, using common portfolio constraints and risk estimates to compare the resulting allocations.

Three benchmarks provide context:

- **SPY:** the passive market benchmark and CAPM market proxy.
- **Equal Weight (EW):** a monthly rebalanced allocation of 1/11 to each sector ETF.
- **Naive MV:** historical mean returns with the same covariance estimator and portfolio constraints as CAPM-MV.

The comparison with Naive MV focuses on the expected-return input. Its value is assessed across return, drawdown, tail risk and trading costs rather than through a single performance measure.

2.2 Universe and source

Monthly adjusted closing prices are sourced from Yahoo Finance for the period January 2007–December 2024. The three years prior to the backtest start date provide the initial rolling estimation window. The full universe is listed in Table 1.

Table 1. Investment universe and approximate S&P 500 sector weights (2024).

Ticker	Sector	Approx. SPY Weight
XLK	Technology	~29%
XLF	Financials	~13%
XLV	Health Care	~12%
XLY	Consumer Discretionary	~11%
XLI	Industrials	~9%
XLC	Communication Services	~9%
XLP	Consumer Staples	~6%
XLE	Energy	~4%
XLU	Utilities	~2%
XLRE	Real Estate	~2%
XLB	Materials	~2%
SPY	S&P 500 (benchmark / market proxy)	—

2.3 Return calculation

Asset returns are calculated as monthly log returns:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right). \quad (1)$$

Here $P_{i,t}$ is the adjusted month-end closing price. The 13-week US T-bill yield ($\text{\textasciitilde IRX}$) from Yahoo Finance is converted to a monthly decimal rate, and the estimation input is $r_{i,t}^e = r_{i,t} - r_{f,t}$.

2.4 Data quality

XLC was launched in June 2018 and XLRE in October 2015. Before launch, the implementation creates synthetic prices by multiplying SPY prices by the ratio of the ETF's launch price to SPY's price on that date. The constant scaling cancels in log returns, so wholly backfilled windows inherit SPY returns and a beta of 1.0. These histories are proxies, not returns on tradable pre-launch ETFs.

Adjusted closing prices account for splits and dividends. The completed analysis series has no material gaps in the backtest window; pre-launch observations use the proxy construction described above.

3 Methodology

3.1 Rolling CAPM expected returns

For each asset i and rebalance date t , a rolling OLS regression is estimated over a 36-month lookback window:

$$r_{i,s}^e = \alpha_i + \beta_{i,t} r_{m,s}^e + \varepsilon_{i,s}, \quad s = t - 36, \dots, t - 1, \quad (2)$$

where $r_{m,s}^e$ is the excess return on SPY at month s . The estimated beta $\hat{\beta}_{i,t}$ captures the asset's current systematic exposure to the market. Expected returns are then constructed as:

$$\mu_{i,t} = r_{f,t} + \hat{\beta}_{i,t} \cdot \frac{\text{ERP}}{12}, \quad (3)$$

where the equity risk premium ERP is set to 5.5% per annum and converted to monthly units, consistent with long-run US estimates [5].

The vector $\boldsymbol{\mu}_t = (\mu_{1,t}, \dots, \mu_{11,t})^\top$ is a model-implied return input. Its cross-sectional differences come from beta estimates and the fixed ERP. This replaces separate historical mean estimates with a common market-risk structure.

Beta estimates are winsorised at the 5th/95th percentile of the cross-sectional distribution each month to prevent outlier betas from distorting the return vector.

3.2 Ledoit–Wolf covariance shrinkage

The sample covariance $\mathbf{S}_t$ computed from 36 months of returns is prone to overfitting: extreme eigenvalues are inflated, producing ill-conditioned inverses that destabilise the optimiser. Ledoit and Wolf [4] propose an analytical shrinkage estimator that convexly combines the sample covariance with a structured target $\mathbf{T}$:

$$\hat{\Sigma}_t = (1 - \delta) \mathbf{S}_t + \delta \mathbf{T}, \quad (4)$$

where $\delta \in [0, 1]$ is an analytically estimated shrinkage intensity. The scikit-learn `LedoitWolf` implementation uses $\mathbf{T} = \bar{v} \mathbf{I}$, with $\bar{v} = \text{tr}(\mathbf{S}_t)/11$: an identity target scaled by average variance [9]. Shrinkage regularises the covariance estimate used by the optimiser.

3.3 Constrained mean–variance optimisation

At each rebalance date, portfolio weights $\mathbf{w} = (w_1, \dots, w_{11})^\top$ are chosen to maximise risk-adjusted return subject to practical constraints:

$$\max_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\mu}_t - \frac{\lambda}{2} \mathbf{w}^\top \hat{\Sigma}_t \mathbf{w} - c \|\mathbf{w} - \mathbf{w}_{t-1}\|_1 \quad (5)$$

subject to:

$$\begin{aligned} \mathbf{w}^\top \mathbf{1} &= 1, & (\text{full investment}) \\ w_i &\geq 0.02, \quad \forall i, & (\text{min. 2\% per sector}) \\ w_i &\leq 0.40, \quad \forall i, & (\text{max. 40\% per sector}) \\ \|\mathbf{w} - \mathbf{w}_{t-1}\|_1 &\leq \tau. & (\text{turnover cap}) \end{aligned}$$

The turnover cap is $\tau = 0.30$ per monthly rebalance and the base risk-aversion parameter is $\lambda = 1.5$. Expected returns and covariance are both expressed in monthly units. The penalty coefficient $c = 0.0010$ represents 10 basis points per unit of L1 turnover; it also discourages large changes in portfolio weights.

Before optimisation each month, $\mathbf{w}_{t-1}$ is updated by drifting the prior weights using realised returns (*drift adjustment*), so the turnover constraint correctly penalises intentional rebalancing rather than passive weight drift:

$$\|\mathbf{w} - \mathbf{w}_{t-1}\|_1 = \sum_{i=1}^{11} |w_i - w_{i,t-1}|. \quad (6)$$

The quadratic programme is solved using SLSQP. Across the walk-forward run, the solver converged successfully on every rebalance solve.

3.4 Walk-forward backtesting

The rolling estimation windows end before the return being evaluated. Portfolio weights are updated monthly and applied to the following month's returns. Table 2 summarises the design.

Table 2. Walk-forward backtest structure.

Phase	Period	Detail
Initialisation	Jan 2007 – Jan 2010	Burn-in: populate 36-month estimation window
Backtest period	Mar 2010 – Dec 2024	178 reported monthly return observations; synthetic histories included
Rebalance frequency	Monthly (end-of-month)	Weights set using only data up to rebalance date
Estimation window	Rolling 36 months	Fixed-length; avoids look-ahead in beta estimation
Cost model	10 bps per unit trade	Deducted from portfolio return on net weight change

At each month-end t : (1) update the rolling return window; (2) estimate $\hat{\beta}_{i,t}$ via OLS for each sector ETF; (3) compute μ_t ; (4) compute $\hat{\Sigma}_t$ via Ledoit–Wolf; (5) drift prior weights by realised returns; (6) solve the constrained quadratic programme; (7) record net trades and costs; (8) apply weights to compute the next month’s portfolio return.

3.5 Performance measures

Metrics are computed over the full backtest period and separately over sub-periods (2010–2014, 2015–2019, 2020–2024) to assess consistency across regimes.

3.5.1 Return-to-volatility ratio (reported Sharpe)

The reported Sharpe ratio is computed from monthly net portfolio log returns as:

$$\text{SR} = \frac{\bar{r}_p \times 12}{\sigma_p \sqrt{12}}, \quad (7)$$

where $\bar{r}_p$ is the mean monthly net portfolio return and σ_p is monthly return volatility. The tables and charts use a zero-risk-free convention for the reported Sharpe measure, so the ratio compares annualised net return with annualised volatility. The risk-free rate enters separately in the CAPM expected-return calculation.

3.5.2 Maximum drawdown

Drawdown is measured using the negative drawdown convention:

$$\text{DD}_t = \frac{V_t - \max_{s \leq t} V_s}{\max_{s \leq t} V_s}, \quad (8)$$

where V_t is cumulative portfolio value at time t . Maximum drawdown is then:

$$\text{MDD} = \min_t \text{DD}_t. \quad (9)$$

Drawdowns are expressed as negative percentages, with more negative values indicating a larger fall from the previous peak. The same measure is applied to each strategy's recorded wealth series.

3.5.3 Tracking error and information ratio

$$\text{TE} = \sigma(r_{p,t} - r_{\text{SPY},t}) \times \sqrt{12}, \quad \text{IR} = \frac{R_p - R_{\text{SPY}}}{\text{TE}}. \quad (10)$$

3.5.4 Calmar ratio

$$\text{Calmar} = \frac{R_p}{|\text{MDD}|}. \quad (11)$$

3.5.5 Turnover

Monthly L1 turnover is:

$$\text{Turnover}_t = \sum_{i=1}^{11} |w_{i,t} - w_{i,t-1}^*|, \quad (12)$$

where $w_{i,t-1}^*$ is the drifted prior weight. Net-of-cost returns deduct 10 bps per unit of L1 turnover per month. This convention is consistent with the turnover cap and transaction cost penalty used in the optimiser.

3.5.6 Historical VaR and Expected Shortfall

Tail risk is measured using one-month historical Value at Risk (VaR) and Expected Shortfall (ES) at the 95% confidence level. Both are reported using the positive-loss convention:

$$\text{VaR}_{0.95} = -q_{0.05}(r_{p,t}), \quad (13)$$

where $q_{0.05}(r_{p,t})$ is the empirical 5th percentile of monthly portfolio returns. Expected Shortfall is the average loss conditional on being in the VaR tail:

$$\text{ES}_{0.95} = -\mathbb{E}[r_{p,t} \mid r_{p,t} \leq q_{0.05}(r_{p,t})]. \quad (14)$$

Because the backtest uses monthly log returns, the VaR and ES figures should be interpreted as historical one-month log-return tail-loss estimates.

4 Results and discussion

4.1 Summary performance

Table 3 summarises backtest performance across all four strategies, net of 10 bps transaction costs.

Table 3. Reported backtest performance, March 2010 – December 2024 realised returns. Net of 10 bps costs. CAPM-MV column shown in bold.

Metric	CAPM-MV	Naive MV	Equal Wt.	SPY
Ann. Return (net)	11.47%	12.23%	11.16%	13.07%
Ann. Volatility	15.53%	14.93%	14.72%	14.84%
Sharpe Ratio	0.739	0.819	0.758	0.881
Max Drawdown	−25.7%	−30.7%	−24.6%	−25.6%
1M VaR 95%	7.17%	7.12%	6.05%	6.61%
1M ES 95%	9.99%	9.18%	9.51%	9.19%
Calmar Ratio	0.447	0.399	0.454	0.511
Ann. L1 Turnover	4.1%	—	≈0%	0%
IR vs SPY	−0.73	−0.21	−0.66	—

CAPM-MV has a reported ratio of 0.739, below SPY (0.881) and Naive MV (0.819) over the same full period. Its maximum drawdown of -25.7% is five percentage points less severe than Naive MV's -30.7% , and its Calmar ratio is higher. Reported annual L1 turnover is 4.1%. However, one-month 95% VaR is 7.17% and ES is 9.99%, so the drawdown advantage does not translate into uniformly lower monthly tail losses.

4.2 Cumulative returns and drawdown

Figure 1 shows the recorded growth of \$1 across the four strategies. The paths are similar through 2018 and diverge more thereafter. CAPM-MV reaches approximately \$4.8, compared with \$6.0 for SPY. Its full-period maximum drawdown is less severe than Naive MV's, but slightly more severe than SPY's.

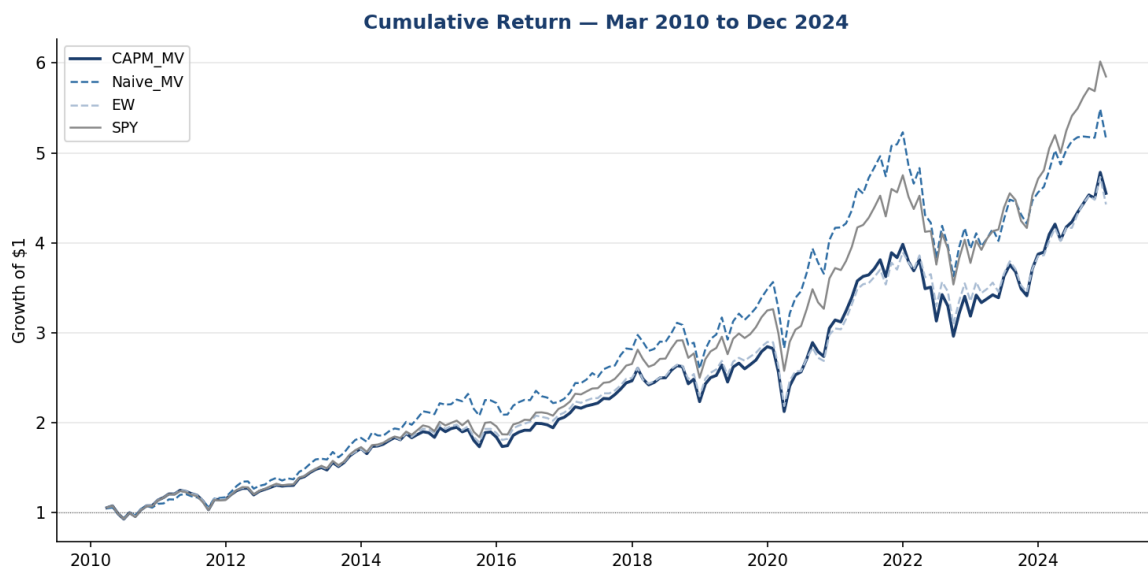


Figure 1. Cumulative return (growth of \$1), March 2010 – December 2024. All realised returns are net of 10 bps transaction costs.

Figure 2 plots drawdown through time with stress episodes shaded. The 2022 rate-shock episode (right shaded region) is particularly informative: CAPM-MV's drawdown is materially shallower than Naive MV, which reached -30.7% at its trough. The COVID crash (left shaded region) shows the opposite pattern—CAPM-MV's beta-structured tilt into cyclicals amplified the drawdown relative to SPY.

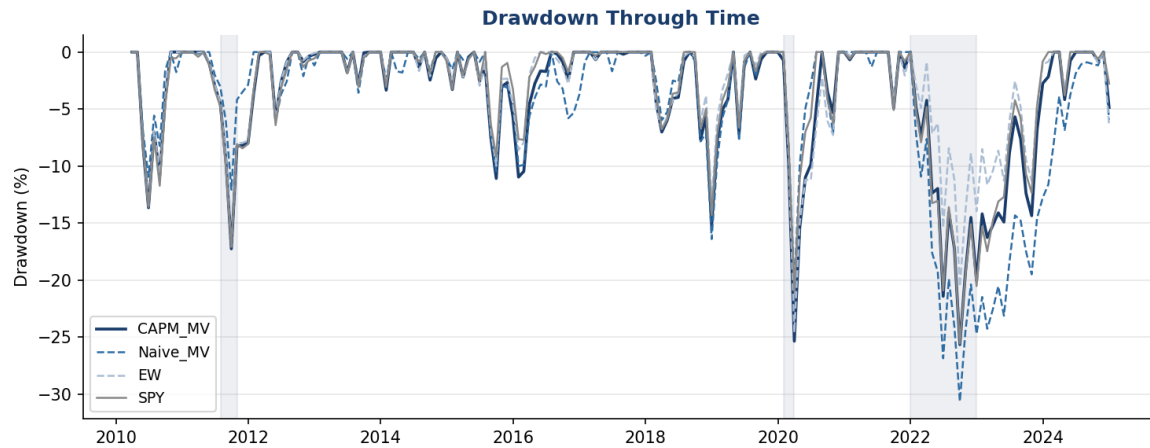


Figure 2. Drawdown through time. Shaded regions indicate the three stress scenarios analysed in Section 4.6.

4.3 Sector weight dynamics

Figure 3 shows quarterly snapshots of the CAPM-MV weights. XLK rises to around 21% by December 2024, while several sectors approach the 2% floor. The changing allocations reflect the combined effect of expected returns, covariance, position limits and turnover penalties.

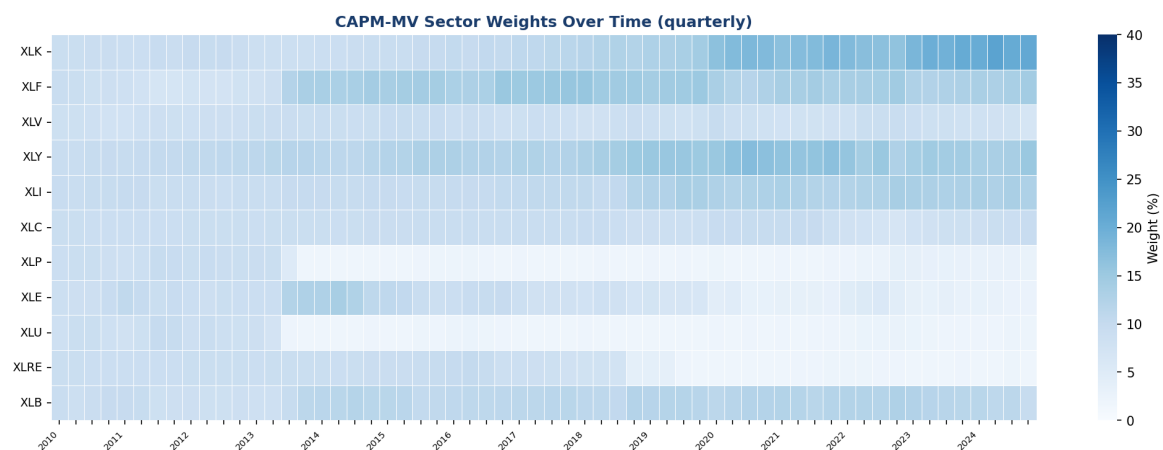


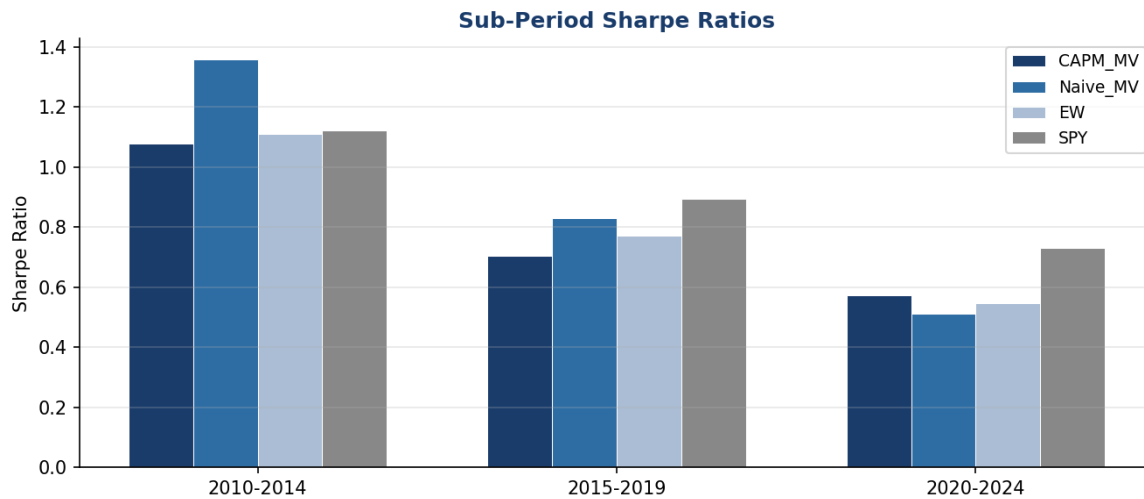
Figure 3. CAPM-MV sector weights over time (quarterly frequency). Darker shading indicates higher weight; white indicates near the 2% minimum.

4.4 Sub-period analysis

Table 4 and Figure 4 break performance down across three five-year sub-periods.

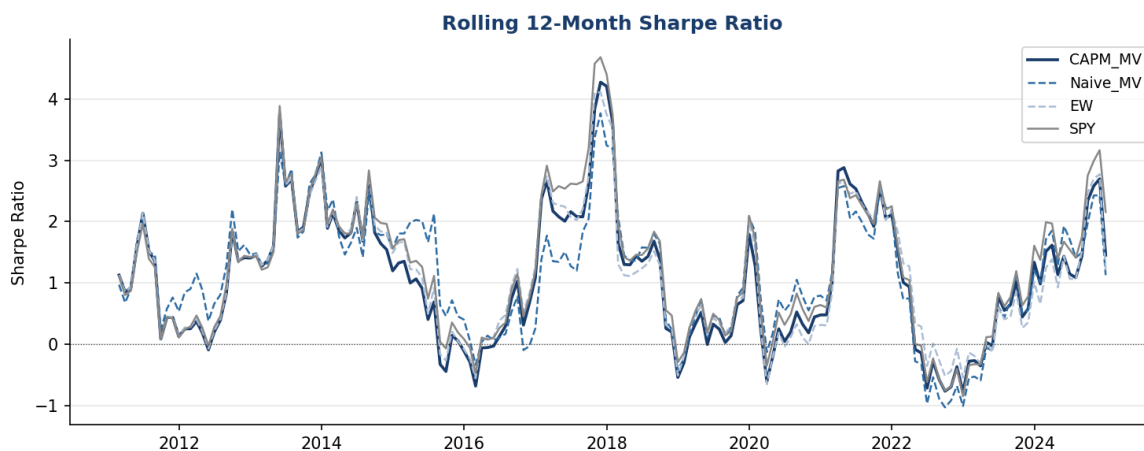
Table 4. Sub-period Sharpe ratios.

Period	CAPM-MV	Naive MV	Equal Wt.	SPY
2010–2014	1.078	1.360	1.110	1.124
2015–2019	0.704	0.832	0.772	0.895
2020–2024	0.575	0.511	0.547	0.731

**Figure 4.** Sub-period Sharpe ratios for all four strategies.

The ranking changes across sub-periods. CAPM-MV trails Naive MV by approximately 0.28 ratio points in 2010–2014 and 0.13 in 2015–2019, then leads by approximately 0.06 in 2020–2024. This variation shows why a full-period comparison is more informative when accompanied by results from distinct market conditions.

Figure 5 plots the rolling 12-month reported Sharpe measure. The strategies alternate in relative performance, reinforcing the importance of examining sub-periods rather than relying only on full-sample averages.

**Figure 5.** Rolling 12-month Sharpe ratio for all four strategies.

4.5 Historical VaR and Expected Shortfall

Table 5 and Figure 6 report one-month historical 95% VaR and ES for each strategy, using the positive-loss convention. The values are deliberately interpreted as empirical diagnostics rather than parametric tail forecasts: with 178 monthly observations, the 95% ES estimate is driven by roughly the worst nine months.

Table 5. Historical one-month 95% VaR and Expected Shortfall. Values are positive loss percentages.

Strategy	1M VaR 95%	1M ES 95%	Worst Month
CAPM-MV	7.17%	9.99%	−17.17%
Naive MV	7.12%	9.18%	−13.45%
EW	6.05%	9.51%	−16.83%
SPY	6.61%	9.19%	−13.92%

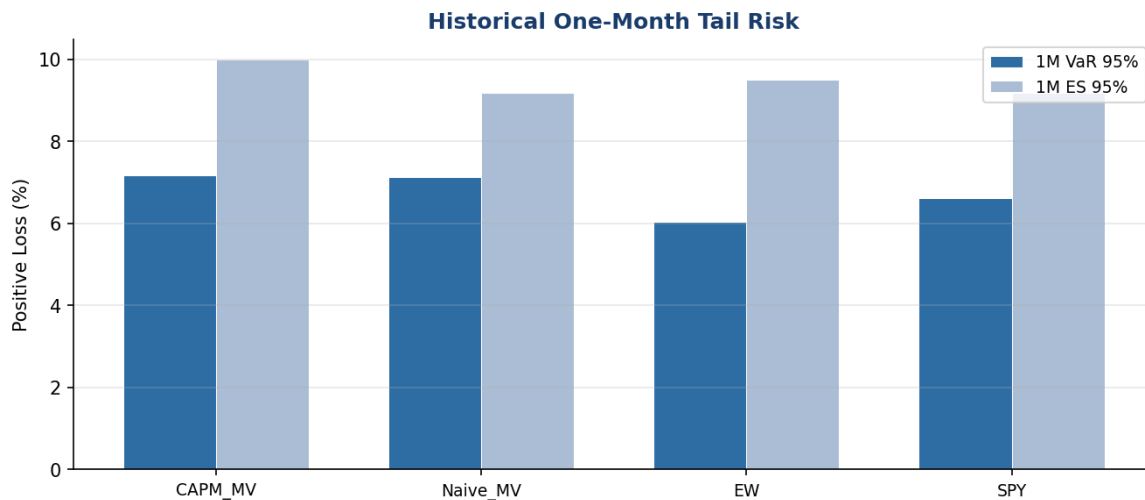


Figure 6. Historical one-month 95% VaR and Expected Shortfall for all four strategies. Values are shown as positive loss percentages.

CAPM-MV and Naive MV have similar one-month VaR at approximately 7.2%, but CAPM-MV has a higher ES of 9.99% compared with 9.18% for Naive MV. This does not contradict CAPM-MV's superior maximum drawdown: maximum drawdown is path-dependent, whereas ES averages the worst monthly returns. The result shows that CAPM-MV reduced the depth of the 2022 drawdown relative to Naive MV, but its worst individual months, particularly around the COVID crash, were more severe.

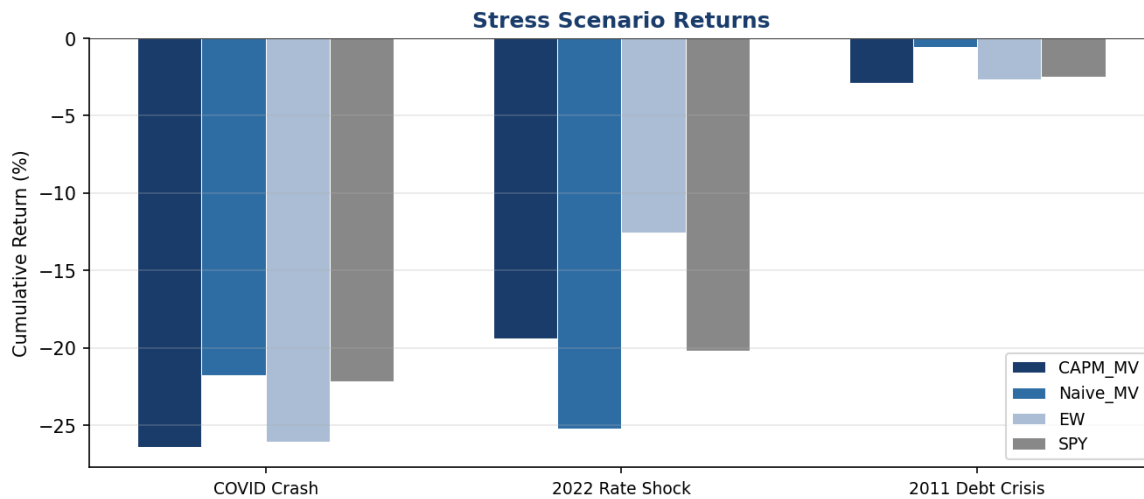
4.6 Stress testing and sensitivity analysis

4.6.1 Historical stress scenarios

Table 6 and Figure 7 report cumulative returns across three historical drawdown episodes.

Table 6. Cumulative returns (%) during stress scenarios. All figures net of costs.

Scenario	Period	CAPM-MV	Naive MV	EW	SPY
COVID Crash	Feb–Mar 2020	−26.4	−21.7	−26.1	−22.2
2022 Rate Shock	Jan–Dec 2022	−19.4	−25.2	−12.5	−20.2
2011 Debt Crisis	Aug–Oct 2011	−2.9	−0.6	−2.6	−2.5

**Figure 7.** Cumulative returns (%) across the three stress scenarios.

The stress windows show different trade-offs. During the COVID window, CAPM-MV performs similarly to Equal Weight and worse than SPY and Naive MV. During the 2022 rate-shock window, it returns −19.4% versus −25.2% for Naive MV, a 5.8 percentage-point advantage; Equal Weight performs best at −12.5%. These differences are consistent with different sector exposures, but a separate attribution analysis would be needed to establish the mechanism. In the shorter 2011 window, all reported losses are smaller.

4.6.2 Parameter sensitivity analysis

Figure 8 shows Sharpe ratios as risk aversion λ and the lookback window are varied one-at-a-time around the base case.

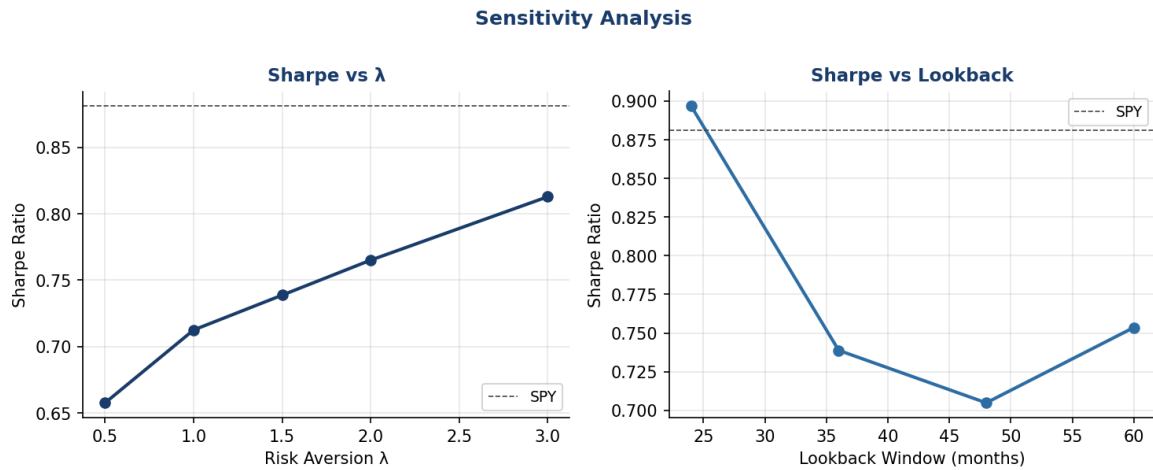


Figure 8. One-at-a-time sensitivity of Sharpe ratio to risk aversion λ (left) and estimation lookback window (right). Dashed line shows SPY Sharpe of 0.881.

Table 7. Sensitivity of Sharpe ratio, maximum drawdown and one-month tail risk to key parameters.

Parameter	Value	Sharpe	Max DD (%)	1M VaR 95%	1M ES 95%
λ sensitivity	0.5	0.658	−29.7	7.33%	10.78%
λ sensitivity	1.0	0.713	−26.8	7.38%	10.34%
λ sensitivity	1.5	0.739	−25.7	7.17%	9.99%
λ sensitivity	2.0	0.765	−26.0	6.90%	9.64%
λ sensitivity	3.0	0.813	−23.4	5.88%	8.78%
Lookback sensitivity	24m	0.897	−26.4	6.94%	9.53%
Lookback sensitivity	36m	0.739	−25.7	7.17%	9.99%
Lookback sensitivity	48m	0.705	−25.6	6.67%	9.83%
Lookback sensitivity	60m	0.754	−24.7	6.34%	9.71%

The ratio increases across the tested λ values. At $\lambda = 3.0$ it reaches 0.813, approaching Naive MV's 0.819, while one-month VaR and ES fall to 5.88% and 8.78%. Within this sample, higher risk aversion shifts the allocation towards a more favourable balance of the measured return and risk outcomes.

The 24-month lookback produces a ratio of 0.897, above the base 36-month result and SPY's 0.881, with one-month VaR and ES of 6.94% and 9.53%. The comparison illustrates how the estimation window influences portfolio behaviour: shorter windows respond more quickly to recent observations, while longer windows draw on a broader history.

4.7 Interpretation of the comparison

The full-period results favour Naive MV on return-to-volatility ratio and CAPM-MV on maximum drawdown. Their relative ranking changes in 2020–2024, illustrating the trade-off between the two return-estimation approaches over this history.

The motivation for structuring return estimates is that a sample mean from 36 observations can be imprecise. Under independent monthly observations its standard error is approximately $\sigma_{\text{monthly}}/\sqrt{36}$. The volatility and mean must be expressed over the same horizon. CAPM replaces direct mean estimation with beta and ERP assumptions; it does not eliminate model uncertainty.

The sector weights and stress-period returns provide context for these differences. A separate attribution analysis could quantify the contributions from sector exposures, beta estimates and changing covariance relationships.

5 Limitations and further work

5.1 Modelling scope

- **Single-factor return model.** CAPM uses market beta and a fixed 5.5% ERP. Multifactor models, such as the Fama–French three-factor model [6], provide an alternative way to represent expected returns.
- **Synthetic ETF histories.** Pre-launch XLC and XLRE observations use scaled SPY prices. These proxy histories support a common analysis window, while approximating the earlier sector exposures.
- **Trading assumptions.** Monthly rebalancing and a fixed cost of 10 bps per unit of L1 turnover simplify execution costs and liquidity conditions.
- **Restricted universe.** The eleven assets are US equity sectors. A broader universe would allow comparisons across asset classes and markets.
- **Tail-risk sample.** Historical 95% ES is based on roughly nine of the 178 monthly observations, so individual stress months have a substantial influence on the estimate.
- **Parameter sensitivity.** Lookback and risk-aversion settings are compared over the same history. Testing them across additional periods would help assess the stability of the observed differences.

5.2 Further work

Useful extensions include broadening the investment universe and comparing the 24-month lookback and $\lambda = 3.0$ settings across additional market periods. The latter reports a ratio of 0.813 and maximum drawdown of -23.4% in the current sensitivity exercise.

Alternative return inputs could include multifactor estimates or Black–Litterman views [7]. A CVaR objective would place greater emphasis on tail losses, while a volatility-based overlay would provide another way to study the interaction between allocation, risk and trading costs.

6 Conclusion

The project implements a rolling CAPM-informed allocation framework and compares it with historical-mean optimisation, equal weighting and SPY. Its main contribution is a transparent comparison of return-input choices under common constraints.

CAPM-MV's reported ratio of 0.739 is below Naive MV's 0.819 and SPY's 0.881. It nevertheless has a less severe maximum drawdown than Naive MV (-25.7% versus -30.7%) and a higher Calmar ratio (0.447 versus 0.399). Its one-month 95% VaR of 7.17% and ES of 9.99% show why drawdown and monthly tail losses should be assessed separately.

The sensitivity analysis extends this comparison by showing how risk aversion and the estimation window affect the allocation. Together, the portfolio weights, stress-period results and tail-risk measures provide a practical framework for assessing the consequences of different modelling choices.

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